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English Script

Slide 1: Title

Good morning everyone. Today we will present our project, Daily arXiv Research Briefing Agent. The goal of this system is to help researchers turn daily arXiv updates into personalized, structured, and evidence-aware research briefings.

Slide 4: Motivation

The motivation comes from a common research problem: arXiv publishes many new papers every day, but manually searching, filtering, and reading them is slow. A researcher may know their topic, keywords, or a few seed papers, but still needs help finding the most relevant new work. Our project addresses this by building an agent workflow that retrieves papers, ranks them according to user interests, and synthesizes the results into a daily research briefing.

Slide 6: Overall System Architecture

The system is organized around one orchestrator and two main skills. The DailyArxivAgentOrchestrator receives user input from the CLI or Streamlit UI, then coordinates the workflow. The first skill is DiscoveryRecommendationSkill, which handles query planning, arXiv retrieval, ranking, and feedback. The second is ResearchSynthesisSkill, which handles extraction, briefing generation, and deep explanation. Together, they produce a daily report and support follow-up results.

Slide 7: Skill 1 Overview

The first skill, Discovery and Recommendation, is responsible for finding and ranking papers. The user can start from interested topics, seed paper, or user

profile established through previous feedback. This skill turns those interest signals into candidate papers and then selects the most relevant ones.

Slide 9: Search Strategy

First, Let's take a closer look at our searching process

If the user provides both a topic and seed papers. The skill will first extract key words from them. And our LLM query planner expands the topic into several controlled arXiv query variants. These variants help to retrieve a broader and more useful candidate pool.

In our system, search is guided not only by the input topic and seed papers, but also by the user profile. The profile stores the user's research interests by saving historical feedback and input seed papers.

If the user provides no topic and no new seed paper. The system can derive search queries from the profile, retrieve papers near the user's long-term interests, and produce a personalized search even without a new topics.

Slide 10: Ranking and Feedback

After retrieval, the ranking stage decides which papers should appear at the top. The system embeds the seed papers, user's profile preference , and candidate papers, then calculate the cosine similarity as the personalized score. The score is then summed with other ranking metrics like category and get the final recommendation ranking .

The user's profile preference can be adjusted by the user's feedback, and further influence future recommendation. If the user likes paper A and dislikes Paper B, then in the future recommendations, papers that are similar to paper A would move up, and papers that are similar to B would move down.

Slide 10: follow-up queries

Another important part of Skill 1 is follow-up queries. After the first recommendation workflow, the system has already stored retrieved papers in the local SQLite database. So when the user asks a follow-up question, such as "Are there any papers about world action model?" the system first searches the locally stored papers. Only when there are no useful local matches, and retrieval is allowed, will it query arXiv again. This makes follow-up interaction faster, more contextual, and closer to how a research assistant should work.

Slide 11: Skill 2 Overview

After Skill 1 selects the papers, Skill 2 explains them. ResearchSynthesisSkill transforms ranked papers into readable research outputs. It can generate a daily briefing for the Top-K papers, and it can also explain one selected paper in different modes, such as method, experiment, or limitations.

Slide 12: Structured Daily Briefing

The daily briefing is not just a plain summary. It contains an executive summary, a summary table, paper-level briefing items, reading priorities, and an evidence boundary. This structure helps users quickly decide what to read first, why each paper matters, and what evidence the system actually used.

Beyond the daily briefing, Skill 2 also supports deep explanation for a selected paper. This explanation is mode-specific: the system does not generate one generic summary. Instead, it downloads the full PDF when available, extracts the paper content, and then generates a structured explanation based on the user's selected angle.

Slide 13: Evidence-Aware Synthesis

A key design principle is evidence awareness. The system distinguishes between metadata, abstract, ranking evidence, and full text. By default, it does not claim to use full-text evidence unless it is actually available for selected-paper explanation, in this way the agent won't make up content. And if an LLM call fails, the system can fall back to metadata-only or deterministic summaries, and it explicitly labels those limitations.

Slide 15: Demo

Now let's demonstrate our two demos.

The first demo takes the “world model” as the topic and two “world action model”-related papers as seed papers. It recommends top-8 relevant papers, all related to the world model. Influenced by the seed paper recommendation, the suggested papers are mostly in the robotics field within the world model framework.

The daily briefing introduces each of the top K papers and explains why each paper is recommended using relevance analysis.

There are also more detailed recommendation diagnostics and worktrace here to check if each component is functioning correctly.

Feedback is provided for adjusting the articles.

You can choose to learn more about a specific article for detailed explanations in a particular direction.

Follow-up queries support more detailed and subsequent searches.

The second demo demonstrates how, without topic or seed paper input, the agent makes recommendations based on your past seed papers and feedback, creating a user profile.

It's noticeable that the recommendations in this scenario highly match my research interests; they are all related to robotics and deep learning, and the recommendation rationale also states that they are highly similar to my research interests.

Slide 16: Evaluation

For evaluation, we use a deterministic offline setup. The test data comes from fixed arXiv-like fixtures, and the LLM is replaced by a fake provider, so the workflow can be tested and reproduced without live arXiv or live LLM calls.

we evaluate multiple workflow levels. The ground truth is manually defined in the fixtures.

Overall, the results show that the workflow behaves as expected in our offline fixture setting. Recommendation and search quality both reach **100%** on the main metrics, meaning relevant papers are retrieved, ranked near the top, and supported with rationales.

Briefing also passes with **100% section coverage** and **100% Top-K coverage**, showing that the report is complete and evidence-aware.

For explanation, method-mode completeness is **0.86** because the fixture intentionally leaves out the corresponding module. This shows the evaluator can detect missing mode-specific fields.

So these results do not claim large-scale benchmark performance. They show the workflow is reproducible, measurable, and structurally reliable under controlled testing.

Slide 18: Conclusion

To conclude, our project builds a complete agent workflow with two public skills. `DiscoveryRecommendationSkill` finds, ranks papers , while `ResearchSynthesisSkill` explains and summarizes them. The agent supports personalized discovery, feedback refinement, follow-up questions, evidence-aware briefing, and reproducible evaluation. In short, it turns daily arXiv overload into a personalized research briefing.